How-To Library

Informatica

Dynamic Data Masking: Capturing the SET QUOTED_IDENTIFER Value in a Microsoft SQL Server or Sybase Database

© Copyright Informatica LLC 1993, 2021. Informatica LLC. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC. All other company and product names may be trade names or trademarks of their respective owners and/or copyrighted materials of such owners.

Abstract

This article describes how to create a Dynamic Data Masking security rule that captures the SET QUOTED_IDENTIFIER value to prevent a database error when you mask queries in a Sybase or Microsoft SQL Server database.

Supported Versions

- Dynamic Data Masking 9.8.1 or later

Table of Contents

Overview. 2
Masking With the AUTH_QUOTED_IDENTIFIER Symbol. 2
Defining the AUTH_QUOTED_IDENTIFIER Symbol. 3

Overview

The SET QUOTED_IDENTIFIER value can cause a database error when you use Dynamic Data Masking to mask data in a Sybase or Microsoft SQL Server database. To prevent database errors, create a security rule that captures the SET QUOTED_IDENTIFIER value and creates the AUTH_QUOTED_IDENTIFIER global symbol.

The SET QUOTED_IDENTIFIER value determines whether the database treats the characters within double quotes as an identifier or as a literal string. When SET QUOTED_IDENTIFIER is ON, identifiers are delimited by double quotes. When SET QUOTED_IDENTIFIER is OFF, identifiers cannot be delimited by double quotes.

To mask statements, Dynamic Data Masking must modify the query based on the SET QUOTED_IDENTIFIER value. For example, the client executes the following statement:

```
set quoted_identifier OFF
```

And then sends the following request to the database:

```
select * from employee;
```

Dynamic Data Masking masks the query and sends the following statement to the database:

```
select   "EMPLOYEE" . "EMPID" , substring("EMPLOYEE" . "EMPNAME" ,1,2)+'**' "EMPNAME"  ,
"EMPLOYEE" . "SAL"
from EMPLOYEE
```

The database returns an error because the masked statement contains identifiers delimited by double quotes. The masked statement requires the SET QUTOED_IDENTIFIER value to be ON.

If the AUTH_QUOTED_IDENTIFIER global symbol exists, Dynamic Data Masking captures the SET QUOTED_IDENTIFIER value and masks the statement in the following way, which does not return a database error:

```
select   EMPLOYEE . EMPID , substring(EMPLOYEE . EMPNAME ,1,2)+'**' EMPNAME  , EMPLOYEE . SAL
from EMPLOYEE
```

Masking With the AUTH_QUOTED_IDENTIFIER Symbol

After you create a security rule that creates the AUTH_QUOTED_IDENTIFIER global symbol, Dynamic Data Masking masks queries based on the SET QUOTED_IDENTIFIER value.

The following examples illustrate how Dynamic Data Masking modifies the queries according to the SET QUOTED_IDENTIFIER value:

SELECT * from EMPLOYEE A

Masked query with SET QUOTED_IDENTIFER OFF:

```
SELECT   A . EMPID , substring(A . EMPNAME ,1,2)+'**' EMPNAME  , A . SAL
from EMPLOYEE A
```

Masked query with SET QUOTED_IDENTIFIER ON:

```
SELECT   A . "EMPID" , substring(A . "EMPNAME" ,1,2)+'**' "EMPNAME"  , A . "SAL"
from EMPLOYEE A
```

Note that in the masked query, Dynamic Data Masking preserves the original identifier `A` as it is set by the user in the original statement.

SELECT * from EMPLOYEE

Masked query with SET QUOTED_IDENTIFER OFF:

```
select   EMPLOYEE . EMPID , substring(EMPLOYEE . EMPNAME ,1,2)+'**' EMPNAME  ,
EMPLOYEE . SAL
from EMPLOYEE
```

Masked query with SET QUOTED_IDENTIFIER ON:

```
select   "EMPLOYEE" . "EMPID" , substring("EMPLOYEE" . "EMPNAME" ,1,2)+'**' "EMPNAME"  ,
"EMPLOYEE" . "SAL"
from EMPLOYEE
```

Defining the AUTH_QUOTED_IDENTIFIER Symbol

Create a security rule at the beginning of the rule set that captures the SET QUOTED_IDENTIFIER value and creates the AUTH_QUOTED_IDENTIFIER global symbol.

The security rule that defines the AUTH_QUOTED_IDENTIFIER symbol must be the first rule in the security rule set. All the rules that define masking for the queries must be after the rule.

After you create and save the security rule, the client must reconnect to the database for the rule to take effect.

Create a security rule with the following properties:

Rule Name

Enter a name for the rule. In this example, the rule name is catch_quoted_identifier.

Rule Matcher

Select the Text matcher. Enter the following regular expression in the Text field:

```
.*\s*SET\s+QUOTED_IDENTIFIER\s+(ON|OFF)\s*.*
```

Select the Regular Expression Identification Method.

The Text matcher identifies database requests that set the QUOTED_IDENTIFIER value. If the request is a match, the rule applies the rule action.

Rule Action

Select Define Symbol. Enter the following parameters for the Define Symbol action:

- Symbol Name: AUTH_QUTOED_IDENTIFIER
- Symbol Value: \(1)
- Keep Per Session: Yes

The rule action creates the AUTH_QUOTED_IDENTIFIER global symbol, which exists for the duration of the client session.

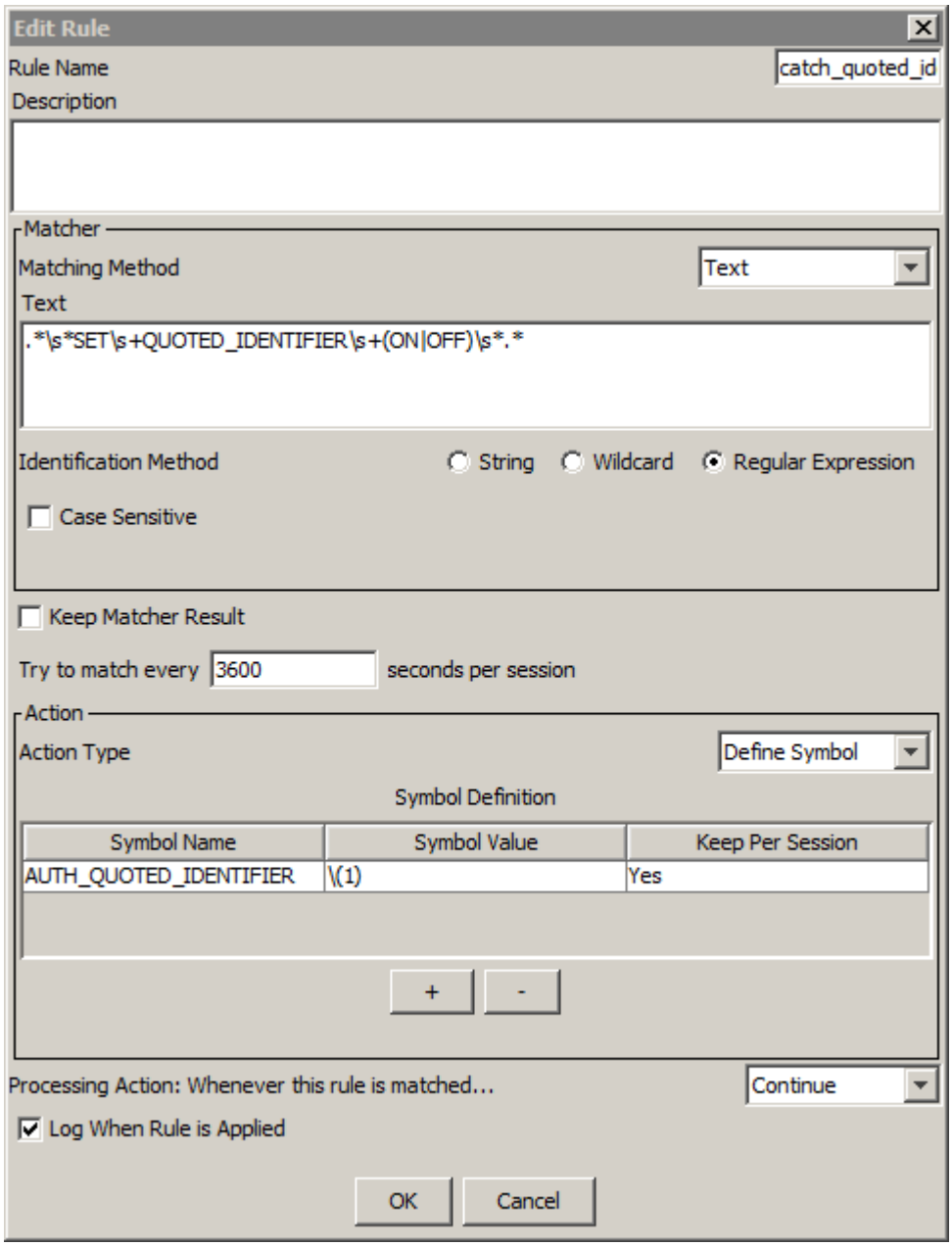
Processing Action

Select Continue. The Continue processing action indicates that the Rule Engine should continue to the next rule in the rule tree.

Log When Rule is Applied

Select the Log When Rule is Applied checkbox to create an entry in the rule.log file if the Rule Engine applies the rule action to the request.

The following image shows the rule parameters:



Save the rule and reconnect the client to the database. Dynamic Data Masking masks queries based on the SET QUTOED_IDENTIFIER value.

Author

Coral Waters

Acknowledgements

Thanks to Arpit Garg for his help in completing this article.